

# Generation Interconnection Combined Feasibility / System Impact Study Report

## For Queue Project Q20-1571

Viele 100 kV  
July 18, 2022

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1571. The Interconnection Customer proposes a 191.5 MW solar photovoltaic facility interconnecting at the Viele 100 kV substation in the Ohio area. The request was received on September 14, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 422 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,434,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                               | Contingency                                | Loading (%) | Rating (MVA) | DFAX (%) |
|--------------------------------------------------|--------------------------------------------|-------------|--------------|----------|
| Swift Crk Chemical Complex – Viele 100kV         | Viele – E Lake Wales 100kV                 | 154.8       | 105          | 100.0    |
| Brisbis – Etiwanda 100kV                         | Peosta – Brownlee Rd 100kV                 | 132.1       | 138          | 6.9      |
| E Lake Wales – Viele 100kV                       | Swift Crk Chemical Complex – Viele 100kV   | 130.5       | 125          | 100.0    |
| Dodge Park Engine No 1 – Edgewater 100kV         | Barrett – Norwich 500kV                    | 126.9       | 119          | 19.7     |
| Swift Crk Chemical Complex – Etiwanda 100kV Line | Peosta – Brownlee Rd Substation 100kV      | 123.4       | 156          | 6.9      |
| A Whoch Substation – Smith Electrical 100kV      | Norwich – Barrett 500kV line               | 109.1       | 150          | 5.5      |
| West Memphis – Brisbois 100kV                    | Barrett – Norwich Substation 500kV line    | 106.0       | 175          | 5.3      |
| Dodge Park Engine No 1 – E Lake Wales 100kV      | Swift Creek Chemical Complex – Viele 100kV | 102.4       | 134          | 100.0    |
| Edgewater – Craigmont 100kV line                 | Barrett – Norwich 500kV                    | 101.0       | 159          | 16.7     |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

#### 4. Required Network Upgrades and Cost Estimates

| Facility                                         | Required Upgrade                            | Cost Estimate | Allocation to Project |
|--------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Swift Creek Chemical Complex – Viele 100kV       | Rebuild line with higher-capacity conductor | \$4,037,000   | 100% (\$4,037,000)    |
| Brisbois – Etiwana 100kV line                    | Rebuild line with higher-capacity conductor | \$4,600,000   | 100% (\$4,600,000)    |
| E Lake Wales – Viele 100kV                       | Rebuild line with higher-capacity conductor | \$1,657,000   | 100% (\$1,657,000)    |
| Edgewater – Dodge Park Engine No 1 100kV L       | Rebuild line with higher-capacity conductor | \$625,000     | 74% (\$461,000)       |
| Swift Crk Chemical Complex – Etiwanda 100kV Line | Rebuild line with higher-capacity conductor | \$3,273,000   | 100% (\$3,273,000)    |
| A Whoch Substation – Smith Electrical 100kV      | Reconductor line                            | \$928,000     | 50% (\$461,000)       |
| West Memphis – Brisbois 100kV                    | Reconductor line                            | \$868,000     | 100% (\$868,000)      |
| Dodge Park Engine No 1 – E Lake Wales 100kV      | Reconductor line                            | \$830,000     | 100% (\$830,000)      |
| Edewater – Craigmot 100kV L                      | Reconductor line                            | \$364,000     | 68% (\$247,000)       |
|                                                  | Total allocated to project                  |               | \$16,434,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.